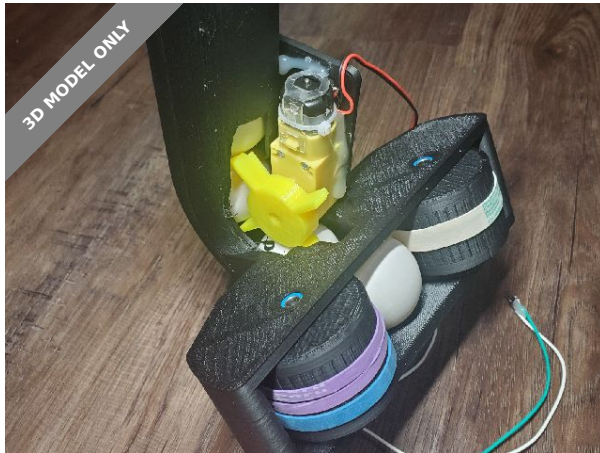




XRP Ping-Pong Ball Shooter



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[VIEW IN BROWSER](#)

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Summary

This is a motorized mechanism allows XRP robots to hold and shoot five ping-pong balls.

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This attachment is a motor powered ping pong ball shooter. It utilizes two flywheels and a feeder motor to push balls through the ball path. It is made to be able to attach to an XRP robot to allow new capabilities with the robot. It is pointed at 30 degree angle to the ground. It can hold and shoot up to 5 balls at a time.

Hardware

The robot uses two 130 size motors to drive the 50mm flywheels at a 1:5 ratio. To feed the balls it uses a TT Motor.

For fasteners I use a combination of hot glue and/or common M3 screws in a variety of lengths.

For the smoothest operation of the flywheels I used 11mm OD and 5mm ID 4mm thick bearings. The flywheels should also be wrapped in rubber bands to get enough grip on the slippery ping-pong balls or else it won't launch.

Operation

The process for shooting balls is simple, just load 5 or 6 into the top of the mechanism, spin up the high speed flywheels, and feed the balls through with the feeder motor.

Printing Instructions

This can be printed in either PLA or PETG. I used 3 shells at 20% infill.

The flywheels, and the ball path should be printed with supports. The rest can be printed without supports.

BOM (Bill of Materials)

TT motor: <https://www.sparkfun.com/products/21245>

130 size DC Motors: <https://a.co/d/bO9bISv>

Bearings: <https://a.co/d/3nHAXYQ>

Any assortment of M3 screws

Assembly Instructions

1. Attach feeder wheel to end of TT motor and fasten the motor to the ball path with either m3 screws or hot glue.
2. Next insert the 11mm OD 5mm ID bearing into the corresponding holes in both shooter sections.
3. Stretch rubber bands over the 50mm flywheels to provide grip on the ping pong balls they should be positioned in the center of the wheels.
4. Then assemble the two sections of the shooter assembly with the 50mm flywheels in the bearings. Fasten the two halves together with 4 m3 screws.
5. With both of the 130 size motors; press the small gear pinions on the end of the shaft. The pinions should have a press fit so they can transmit sufficient torque to the flywheels.
6. Then push the motors into their spots on the shooter sub-assembly so the gears mesh without excessive resistance. You can hot glue them in place
7. With the feeder assembly and the shooter assembly made you can combine them as shown in the pictures with 4 m3 screws.

Model files

flywheel-motor-pinion-v2.step

50mm-flywheel-v5.step

ping-pong-ball-shooter-v29.step

two-motor-flywheel-v42.step

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